

Using hardware devices

After you have successfully created a project, it can be run either on a virtual device (emulator) or on your own device. Though seeing your app run on your own device can be fun, the advantage of running it on the emulator is that it can be verified on different versions of the Android platform. You also have the option to check how it functions in different orientations and screen sizes. To run the app on your device:

- First connect the device to the development machine through a USB cable.
- Go to **Settings > Developer Options** on your device. If you use Android versions higher than 4.2, the developer options are hidden. So, to activate them, go to **Settings > About phone** and tap **Build number** seven times. Then go back to the previous menu where you will now find the **Developer Options**.
- Enable the **Debugging over USB** option.
- Now, go to your project and select one of your project's files and click **Run** from the toolbar.
- In the **Choose device** window that appears, select the **Choose a running device** radio button, select your device, and click **OK**.

Building and running a project

Before you can run your application on the Android emulator, verify the default AVD or create one. To run (or debug) your application, select **Run --> Run (or Run --> debug)** from the Android Studio menu bar. It will compile the project, create a default configuration, install and finally run on an emulator. If you run the application in debug mode, the **Choose a Device** option appears so you can select an attached device or emulator.

Running from the command line

To build in debug mode, open a terminal and navigate to the root of your project directory. Use Gradle to build your project in debug mode, invoke the **assembleDebug** build task using the Gradle wrapper script, as follows.

```
$ chmod +x gradlew
$ ./gradlew assembleDebug
```

To see a list of all the available build tasks for your project, type the following command:

```
$ ./gradlew tasks
```

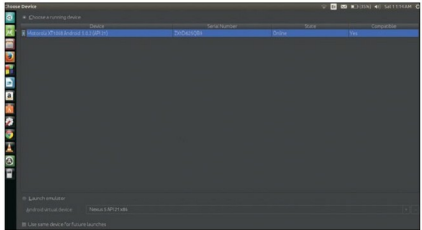


Figure 5: Selecting an external device

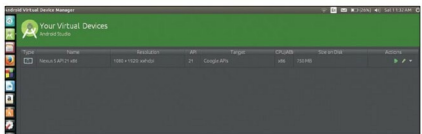


Figure 6: Selecting a virtual device (Nexus 5)

Tips

- Here are few tips for newbies. The following short cuts can save a lot of time and help you work efficiently.
- It would be very easy if when you missed adding attributes to your button or text view, a rendering message gets displayed, which, when clicked upon, would automatically add all the missing attributes.
- While creating a new project/activity, you can select the type of activity you want. It has very smart auto-code completion. Just pressing **Ctrl+space** will enable this feature.
- Method documentation can be obtained by hovering your cursor over the method definition, and clicking and pressing **F1** will bring a pop-up box with the required details.
- Alt+Enter**: This helps in fixing coding errors like missing imports, variable assignments, etc.
- When a reference to an image is inserted in the code, a preview appears in the margin. Pressing **F1** with the preview image selected displays the resource asset details. **END** 

References

- [1] <http://developer.android.com/intl/zh-tw/tools/studio/index.html>

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